

This is a **team** competition. Please form teams of 3, 4 or 5. It runs in two rounds. Everyone competes in Round 1. A few teams advance to Round 2, where you will get new information about the company and have one night to work out what it does to the recommendation you already made.

Round 1. Your team submits a written analysis to Box in PDF format. Four pages of narrative, maximum. Exhibits — tables, figures, regression output — go in an appendix that does not count against the four pages, so do not crush your writing to make room for a table. Put every number you rely on in the appendix and refer to it from the text.

Round 2. Finalist teams receive an additional packet the evening before final presentations and present to a panel of judges. Twelve minutes, then eight minutes of questions. You are not expected to redo Round 1 overnight. You are expected to tell the panel what the new information changes and what it does not.

Some notes on case competitions:

- DO NOT confuse a case with a homework problem. Homework has an answer. A case has a decision, and reasonable people looking at the same data will land in different places. You are being judged on the quality of your argument, not on whether you matched a key.
- **A defended range beats a confident point estimate.** You'll be tempted to report a number — this sounds authoritative and a range can feel like hedging. A team that reports one number with false precision will lose to a team that reports the range and knows why it is a range.
- DO say plainly what you could not figure out. Every real analysis has a hole in it. Naming yours is evidence you understand the problem.
- DO NOT bury the recommendation on page four. Lead with what you think the company should do, then show the work that got you there.
- You can use any online tools you like, including large language models. But you own every number you submit. "The model said so" is not a defense.
- DO NOT be a jerk by letting your more motivated teammates do all the work. DO pull your weight. Divide the analysis, but make sure every member of the team can explain any part of it — questions in Round 2 will not be addressed to the person who happens to know the answer.
- DO NOT submit work with mis-spelled words, poor grammar, or incoherent thoughts. DO edit carefully and submit work that you would be proud to show to a prospective employer.
- DO NOT submit a graph without proper labels. DO label your axes, state your units, and say what the sample is.
- A very good analysis is possible with the regression and causal-inference tools you have already learned. Start there.

Meridian Casualty

Hiring Is Broken — Can You Fix It?

The company. Meridian Casualty is a property and casualty insurer, selling auto, homeowners and small commercial policies in twenty-six states. When a customer files a claim, a claims adjuster in one of Meridian's forty regional service centers investigates and settles it: the adjuster takes the first call, orders the estimate, decides what the policy covers, authorizes payment, and closes the file.

A claim is *reopened* when it requires additional work after an adjuster has closed it. Meridian measures operational performance of adjusters using the reopen rate: the percentage of an adjuster's closed claims that are later reopened. A lower rate is better, and reopenings are not always the adjuster's fault, so the measure is informative rather than perfect.

These are high-volume jobs. A center runs anywhere from roughly one hundred to five hundred people, and Meridian hires roughly twenty new claims adjusters per center per month. Early attrition in claims work is high everywhere in the industry and it is high at Meridian: a meaningful share of every new class is gone within six months. Nobody at Meridian thinks that is good, but for years it was stable and budgeted for.

The problem. It stopped being stable. Meridian's Chief People Officer, Dana Whitfield, has been watching retention-after-six-months slide for roughly two years, and it has now slid far enough that she has taken it to the executive committee. Her description of the problem, in a memo to that committee:

"Two things are happening at once. Turnover in the first six months is up sharply, and it is up most in the centers I would have told you were our best-run. At the same time my recruiters tell me the applications have never looked better. Everyone who applies now looks qualified. So we are seeing stronger applicants than we have ever seen and getting worse hires than we have had in years."

"My recruiters assume that applicants are using AI to write the situational answers on the application. I do not need anyone to tell me that AI exists; instead, I need to know what this is costing us (if anything) and then what (if anything) we should do. Every option I can think of costs real money, several of them make the applicant experience worse, and I cannot tell which one leaves us better off."

Treat AI-assisted writing as the working hypothesis. Your task is not to prove that applicants use AI. It is to determine whether the hiring process has lost useful information, and if so, what that has cost Meridian, and what Meridian should do.

How hiring works at Meridian. Every center first requests a job posting for a specific role. Applications arrive online, and they arrive in enormous numbers: a typical opening draws well over a hundred applications, and in recent years many more than that. Meridian does not have the recruiting staff to read all of them. Most applications are not individually reviewed. A recruiter works down the queue and reviews as many as the week allows.

An application a recruiter reviews contains the usual things — a résumé, work history, whether the applicant holds an active adjuster license, whether anyone at Meridian referred them, whether they are currently working — and one thing that is a little unusual. Meridian’s application includes a mandatory **situational section**: three open-response questions about handling difficult claims situations. “Describe how you would handle a customer whose claim was denied and who believes the denial was a mistake” is a representative one.

Meridian calls this the situational section. Its total score appears in the data as `essay_rubric_score`.

The recruiter scores the answers on a rubric — completeness, specificity, organization, and whether the response shows evidence of care — zero to ten per question, so zero to thirty overall. That score is a real input to the screening decision, and it has been for as long as anyone remembers. Meridian introduced the situational section in the early 2010s because the people who wrote good answers turned out to be the people who lasted, and for a decade the section did what it was supposed to do.

Applicants who clear the screen are interviewed, and interviewers record a structured rating. Meridian makes offers, most are accepted, and new hires start in the same calendar month or within the next three months.

The flex workforce. Meridian also runs a year-round contract staff to absorb volume that its permanent headcount cannot. Catastrophe seasons, a bad quarter for hail, a center that is short-handed — the flex pool covers it. These people are supplied through a staffing partner, and they come out of Meridian’s own applicant flow: they applied for permanent roles at Meridian, were not selected, and were subsequently placed on contract assignments by the partner. The partner places them on certification, geography and availability. Flex staff do the same claims work, in the same centers. Meridian calculates their reopen rates in the same way it does for permanent adjusters.

The two systems. Meridian has been replacing its applicant tracking system — the software that collects applications and routes them to recruiters. The two systems handle the situational section differently.

- On the **legacy** system, the situational section is a timed field inside the browser. The applicant opens it, the clock starts, and the answers have to be typed there in one sitting. Pasting into the field is disabled.
- On the **new** system, the section is untimed. Applicants can start it, leave, come back, compose their answers anywhere they like, and paste the result in when they are ready.

The replacement did not happen in one go. A vendor pilot moved a small group of centers onto the new system in 2021. The rollout then stalled for about eighteen months while Meridian and the vendor renegotiated the contract, and resumed as a full program in 2023 and 2024. Seven centers are still on the legacy system today and have no scheduled migration date. Which centers went in which wave was decided on IT readiness and center size, not at random — the pilot centers were the large ones with the strongest local technology staff.

What you have. Meridian has given you five years of internal data — January 2021 through December 2025, all forty centers. Five files: every application a recruiter reviewed, every permanent hire with first-six-month outcomes where enough time has elapsed, every flex placement and how theirs went, a center-by-month operating panel including the cost rates Meridian’s finance group uses, and the migration date of each center. A data dictionary describes all five, and a starter script loads them and does the joins so that you do not spend your first afternoon on merge keys.

The application-level file contains the *reviewed pool*, not every application Meridian received. Total application volume appears in the center-by-month file.

What you must deliver. Three things.

- (a) **What Meridian’s hiring process gets right, and where it fails.** Not a list of everything you noticed. A judgment about a process, supported by evidence.
- (b) **Whether the situational section still works as a screen, and how you know.** The second half of that sentence is the important part. Be precise about which comparison you are making, on whom, and what would have to be true for it to mean what you say it means.
- (c) **A recommendation, with its cost.** What should Meridian do, what will it cost, and what would have to be different for you to recommend something else.

Do not mistake volume of analysis for quality of argument. Three good answers beat nine partial ones, and a team that hands in eleven exhibits without a recommendation has not done the assignment.

These are the required outputs. Placebos, alternative specifications, bounds, and other robustness work are ways to strengthen those three answers — not additional deliverables.

The cost worksheet. Meridian’s finance group has provided a standard framework for translating your estimates into a cost per retained employee. Here, that means the expected cost required to produce one filled seat for six months.

Suppose the probability that a newly hired employee remains at Meridian for at least six months is r . On average, Meridian must therefore hire $\frac{1}{r}$ people to obtain one retained employee. Of those hires, $\frac{1}{r} - 1$ will separate before reaching six months.

The expected cost per retained employee is therefore

$$\begin{aligned} \text{cost per retained employee} &= \frac{C_{\text{hire}}}{r} + C_{\text{sep}} \left(\frac{1}{r} - 1 \right) \\ &\quad + C_{\text{policy,retained}} + [C_{\text{quality}}]. \end{aligned}$$

The components differ in when they are incurred:

Component	Incurred when
C_{hire}	All costs required to produce one hire under the policy: sourcing, screening, interviews, assessments, training, lost productivity while the employee ramps up, and vacancy cost. If a cost is quoted per applicant, multiply it by the number of applicants who incur it for each hire.
C_{sep}	Each hire who leaves before six months.
$C_{\text{policy,retained}}$	Once, if the employee reaches six months; for example, a six-month retention bonus.
C_{quality}	The operational-quality cost for one retained employee. Multiply claims handled by the predicted reopen rate and the cost of a reopened claim.

Each row of `center_month.csv` describes one center in one month. The worksheet inputs in that file are not costs “per center.” Each has its own unit: dollars per recruiter or grader hour, per opening, per hire, per vacancy day, per separation, or per reopened claim. Claims volume is measured as claims per adjuster over six months. These inputs vary by center and over time. The data do not provide the design or price of a hypothetical policy.

Days to fill is the average number of days from opening a requisition to accepting an offer. During those days, Meridian bears the incremental cost of an unfilled desk through overtime, flex coverage, or work that waits. For each hire, the vacancy component of C_{hire} is

$$\text{days to fill} \times \text{vacancy cost per day}.$$

It remains inside C_{hire}/r when cost is converted to a per-retained-employee basis. This matters because early separations require replacement hires and therefore create additional vacancy spells. A policy that takes longer to fill an opening must produce enough improvement elsewhere to justify those additional vacancy days. Use the observed days to fill to cost the status quo. For a hypothetical policy, state and defend an assumed change in days to fill and show how sensitive the recommendation is to it.

A worked example. Suppose — and these three numbers are invented — that a policy produces a six-month retention probability of $r = 0.75$, thirty days to fill, and a reopen rate of 6%. Using 2025 average rates:

- The cost of each hire is

$$C_{\text{hire}} = 747 \text{ (sourcing)} + 5,085 \text{ (training)} + 6,540 \text{ (ramp)} + 30 \times 219 \text{ (vacancy)} = \$18,942.$$

- Producing one retained employee requires, on average, $1/0.75 = 1.33$ hires. It therefore generates $1.33 - 1 = 0.33$ expected separations.
- Ignoring any additional policy costs, the hiring and separation cost per retained employee is

$$\frac{18,942}{0.75} + 2,540 \left(\frac{1}{0.75} - 1 \right) = 25,256 + 847 = \$26,103.$$

- If operational quality is monetized, its expected cost is

$$609 \text{ (claims)} \times 0.06 \times \$323 \approx \$11,800$$

per retained employee.

The example highlights two important points. First, the operational-quality term is large enough that including or excluding it could change which policy is preferred. Whether to monetize operational quality is a judgment you should make explicitly, not implicitly by omission. Second, a policy that improves retention but increases days to fill trades fewer separations against greater vacancy costs. You therefore cannot rank the policies on retention alone.

The decision. Producing good information about applicants is not free. Somebody has to bear that cost, and Meridian's realistic options are:

- (a) **Do nothing.** Tolerate the current process. Always on the table and never automatically wrong.
- (b) **Make applicants bear it.** A longer, timed, proctored or otherwise costly application. More informative per applicant, and fewer applicants.
- (c) **Make Meridian bear it.** Structured interviews, work samples, paid work trials, more recruiter hours per opening. Expensive, and the expense falls on the party that can afford it.
- (d) **Bear it after hiring.** Accept a noisy screen and sort people once they are on the job — a real probationary period, faster termination decisions, retention incentives.

Recommend one. Cost it against the status quo or another credible alternative. You are not required to cost every possible implementation. Tell Whitfield what would have to be true for you to have recommended something different.

About Round 2. If your team advances, you will receive a short packet from Meridian containing information you do not have now. It will not require you to rebuild anything. It will require you to say what it does to the recommendation you made in Round 1 — which may be nothing, and if the honest answer is nothing, say that and defend it.